

Week 7 executive brief: control cost before scaling

Decision requested: approve an educational demo and a gated API pilot plan. **No-go:** clinical use or a self-hosted GPU purchase without further evidence.

The service turns a short symptom message into a structured demo urgency response. It is **not medical advice**. This project demonstrates cost controls and reliable software behaviour using a free deterministic substitute for an AI model.

Monthly planning scenario	Before optimisation	With 60% exact-cache hits
100,000 requests	\$147.55 total; \$1.4755/1,000	\$137.02 total; \$1.3702/1,000
Sustained 10×: 1,000,000 requests	\$305.50 total; \$0.3055/1,000	\$200.20 total; \$0.2002/1,000

These are modelled costs, not bills. They include \$30/month hosting and four operations hours at \$25/hour. Token rates, message lengths, traffic and staffing are visible assumptions. The artificial fixture's 60% hit rate is not a demand forecast.

Lever 1 — exact caching: reuse an identical message's response for up to ten minutes. In the measured 100-request fixture, model items fell from 100 to 40. This models 60% token savings, but only **7.14% total savings at baseline** because staffing and hosting remain. Keys preserve wording and punctuation; no semantic matching is used.

Lever 2 — batching: group up to ten waiting requests. After caching, measured provider calls fell from 40 to 4; model items stayed at 40. **No additional dollar saving is claimed.** Small queues may add waiting time. It is a transport efficiency result from the stub, not a paid-provider discount or GPU benchmark.

Performance and quality: the recorded local sweep completed 100/100 contract checks in every configuration. p95 was 24.45ms baseline, 25.33ms cached, and 25.53ms cached plus batching. p95 means 95 out of 100 requests completed within that time. The demo objective is $p95 \leq 100ms$; these short stub timings do not predict live AI latency. Quality here means preserving expected synthetic responses, not proving clinical safety.

Primary risk: a cheap response may still be clinically wrong. Mitigation: keep the service educational; require clinician-reviewed evaluation and privacy approval before any patient use. In-memory caching and queues are bounded but volatile. Real deployments need tenant isolation, durable spend accounting and reviewed retention; this demonstration uses no patient data.

Budget protection: a simulated \$1/hour application guard refuses model work when its reservation ceiling is exhausted. AWS budget examples alert at 80% actual and 100% forecast of \$250/month; alerts do not stop spend and have not been deployed. External API invoices and people time require separate reconciliation. Unoptimised 10× traffic exceeds the overall \$250 planning envelope.

API versus self-host: prefer API for a controlled pilot. The estimated cost crossing is 9.32m requests/month without caching, 23.31m with equal 60% caching on both sides. GPU capacity and model quality are untested; baseline demand is far below either scenario. See the memo for the equation and staffing assumptions.

Conditions to proceed: verify current prices and realistic demand; independently validate quality; set live request/token/spend limits; prove alert receipt and shared spend enforcement; assign a budget owner. No paid-provider experiment or cloud production deployment has occurred. Evidence: *levers/measurements.md*, *cost/cost_model.md*, *memo/api_vs_selfhost.md* and *evidence/EVIDENCE_CHECKLIST.md*.